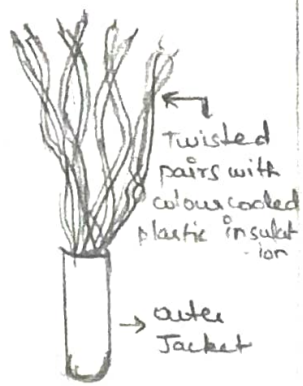
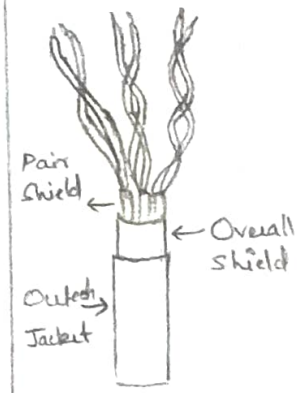
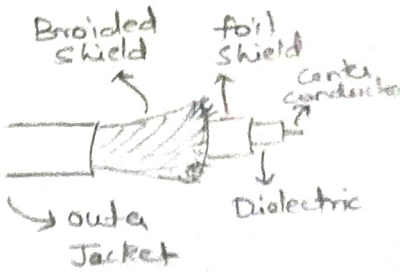
Exp : 2 Date 30/4/24	Study of different types of Network Cables.

a. Understand different types of network cables

Different type of cables used in networking are:

1. Unshield Twisted Pair (UTP) Cable
2. Shield Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable Type	Category	Maximum Data Transmission	Advantage / Disadvantage	Application / Use	Image
UTP	Category 3	10 bps	**Advantages:** * cheaper in cost * Easy to install as they have a smaller overall diameter **Disadvantages:** * More prone to (EMI) Electromagnetic interference and and noise	10Base - T Ethernet	
	Category 5	Upto 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6, 6a	10 Gbps	**Advantages** * shielded * Faster than UTP * less susceptible to noise and interference **Disadvantages** * Expensive * Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres.	
SSTP	Category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (100 m)	

Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	**Advantages** * High bandwidth * Faster than UTP * Less susceptible to noice and interference * versatile **Disadvantages** * Limited distance * Cost * Size is bulky	Speed of signal is room television network High speed internet connections	
fibre Optics Cable	Single mode Multi mode	100 Gbps	**Advantages** * High speed * High bandwidth * High security * Long distance **Disadvantage** * Expensive * Requires skilled installers	Maximum distance of fibre optics cable is around 100 meters.	

Result:

The types of network cables and making cross wired cable & straight through cable using clamping / crimping tool is studied.